Imad Serhal

Mechanical & Control Systems Engineer | Industrial Automation

Email: imadserhal7@gmail.com | Phone: +39 345 509 5258 | Location: Italy
LinkedIn: linkedin.com/in/imadserhal | Portfolio: imadhserhal.github.io

PROFILE

Engineer with a B.Eng. in Mechanical Engineering and an M.Sc. in Automation & Control Systems Engineering (University of Padova). At STIGA I carry out tests on electric and autonomous machines from start to finish, from bench setup to the issued report. Before that I designed HVAC and pumping systems for residential buildings in Lebanon. My M.Sc. thesis developed the control of an electrically excited synchronous motor for a heavy-duty IVECO vehicle, and I have programmed Siemens PLCs in TIA Portal. Open to roles in mechanical engineering, control systems and industrial automation.

TECHNICAL SKILLS

Mechanical & Thermal: AutoCAD, SolidWorks, HVAC and hydronic sizing, thermal load calculations, pump and fan sizing, Bills of Quantities, shop and as-built drawings, thermodynamics, fluid mechanics, heat transfer

Control Design: PID/PI control, cascaded loops, anti-windup, state-space control, LQR, observers, Kalman filtering, digital control, Extremum Seeking Control

Electric Drives: Field-Oriented Control (FOC), MTPA, flux weakening, MTPV, d-q motor models, flux and torque maps

Modelling & Simulation: MATLAB, Simulink, Simscape, nonlinear simulation, system identification, lookup tables

PLC & Automation: Siemens TIA Portal V15 (IEC 61131-3), S7-PLCSIM, Factory I/O, sequential logic and interlocks, analog process control

Testing & Measurement: Test procedures and reports, acceptance criteria, Hioki data loggers, thermocouple instrumentation, climate chamber testing, CAN bus (J1939, CANopen) log analysis, EN 50636, ISO 5395-3

Software: Python (PySide6, pandas, NumPy, matplotlib), C/C++ (STM32, Arduino), REST APIs

PROFESSIONAL EXPERIENCE

Control & Testing Engineer Apr 2025 - Present
STIGA Italy

- Carried out 25+ tests from start to finish: set up the test benches and instruments, collected and analysed the data, and wrote and issued the reports, each with procedure, acceptance criteria, results and a pass/fail verdict.
- Wrote the functional validation checklists (based on EN 50636) for an autonomous robot's firmware and app, and used them for acceptance checks of firmware releases, both production and over-the-air updates.
- Verified electrical and electronic components against standards and internal norms: PCB vibration tests, DC-DC converter thermal and protection tests (OCP, OVP, UVP), emergency-stop function and brakes to ISO 5395-3.
- Measured voltage, current, power and temperature with Hioki data loggers and read motor speed, bus current and fault codes from CAN-bus logs to find root causes and support firmware improvements.
- Integrate and debug motors, motor controllers, batteries, sensors, LiDAR and NVIDIA Jetson computers on prototype machines.
- Built a Python desktop tool on my own initiative that turns data-logger and CAN-bus logs into a finished Word test report.

Research Trainee, Advanced Motor Control Feb 2024 - Jun 2024
University of Padova Italy

- Built linear and nonlinear control-oriented models of IPM, SPM and EESM machines, including excitation dynamics, saturation and cross-coupling.
- Designed and tuned cascaded current and speed PI loops with anti-windup and bandwidth separation.
- Generated flux, torque and speed maps to evaluate operating envelopes and controller performance.

Mechanical Design Engineer Jun 2021 - Sep 2022

My Projects SAL Lebanon

- Designed HVAC, fluid and pumping systems, with thermal load calculations and sizing of circulation pumps, compressors, fans and heat exchangers.
- Prepared Bills of Quantities and produced AutoCAD shop drawings and as-built drawings for documentation and client handover.
- Supervised the installation of mechanical equipment on active construction sites, checking compliance with the design specifications and safe operation.

PROJECTS

M.Sc. Thesis: Control of an Electrically Excited Synchronous Motor (IVECO heavy-duty EV)

- Designed the full control architecture in MATLAB/Simulink: field-oriented control with MTPA, flux weakening and MTPV, lookup tables and the switching logic between operating regions.
- Applied Extremum Seeking Control to the excitation current, cutting simulated Joule losses by up to 10%.

PLC Automation: three virtual-commissioning cells (Siemens TIA Portal V15, S7-PLCSIM, Factory I/O)

- Conveyor sorting by height, multi-level elevator with interlocks, and tank water-level control with an analog level transmitter and valves.
- Each program verified against a simulated 3D plant through S7-PLCSIM. Demo videos at imadhserhal.github.io.

Ethanol-diesel dual-fuel engine study (B.Eng. final year project)

- Tested E0 to E30 blends on a single-cylinder diesel test bed, built the performance calculation sheet and checked a Diesel-RK model against the measurements.

EDUCATION

M.Sc. Automation & Control Systems Engineering 2022 - 2024

University of Padova, Italy

B.Eng. Mechanical Engineering 2017 - 2022

Beirut Arab University, Lebanon

LANGUAGES

English: C1 (Fluent) | Arabic: Native | Italian: A2 (Improving)